

Dynamic Host Configuration Protocol

- Topics covered:
 - The Purpose of DHCP
 - Basic Functions of DHCP
 - Configuring DHCP in Cisco IOS

The Purpose of DHCP

DHCP Overview and Purpose

DHCP is used to allow hosts to **automatically and dynamically learn network configuration**, including:

1. IP address
2. Subnet mask
3. Default gateway
4. DNS server

This removes the need for **manual and static configuration**.

In practical use:

- When a device connects to WiFi or a network, it automatically gains access
- Users do not need to ask for IP details or configure anything manually

This is why DHCP is considered an **essential part of modern networks**.

Usage distinction:

- Client devices (PCs, phones) → use DHCP
- Routers and servers → usually manually configured (require fixed IPs)

Network size behaviour:

- Small networks → router acts as DHCP server
- Large networks → dedicated Windows/Linux DHCP server

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Verifying DHCP Configuration (Windows GUI)

Now I will give a brief demonstration of DHCP's function using my own PC, which is running Windows 10. This is 'network and sharing centre' which can be accessed from control panel

Navigation:

Control Panel

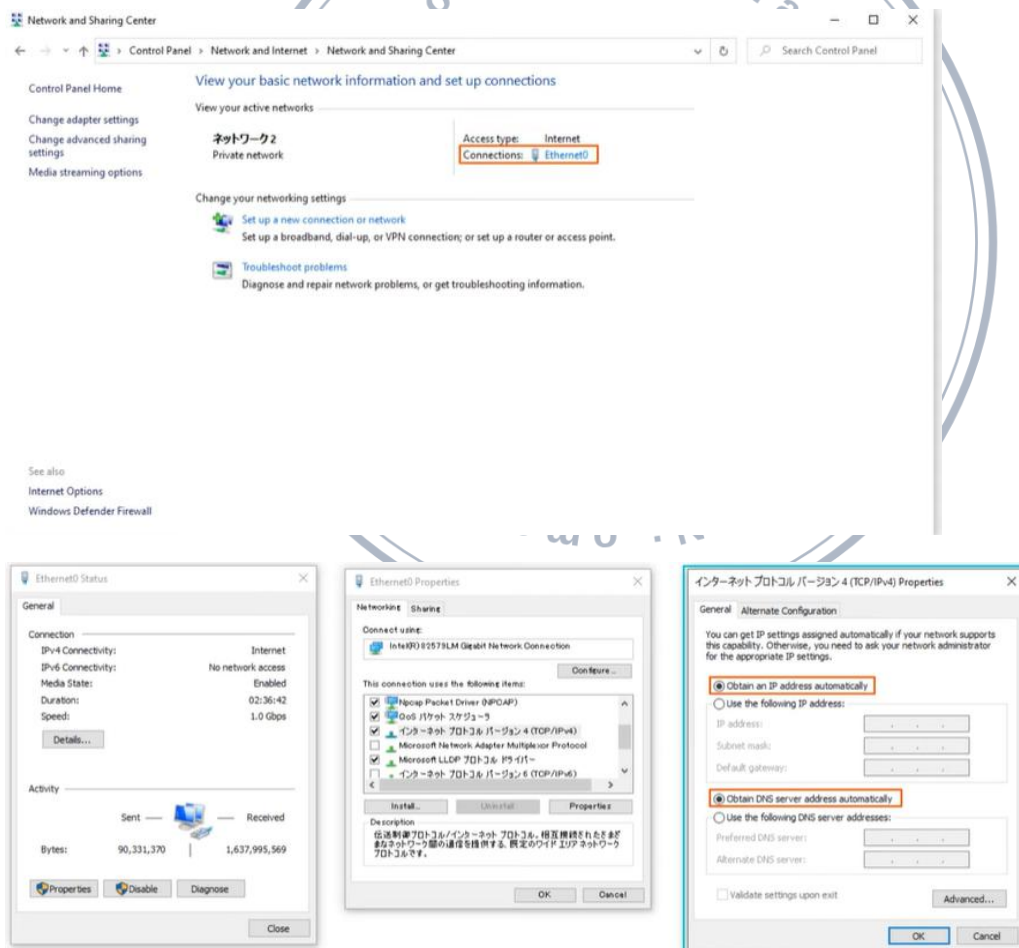
→ Network and Internet

→ Network and Sharing Center

→ Ethernet0

→ Properties

→ IPv4 Properties



Settings:

Obtain an IP address automatically

Obtain DNS server address automatically

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Explanation:

These settings confirm that the PC is using DHCP.

- “Automatically” means the PC will **learn all network settings dynamically**
- No manual IP or DNS configuration is required

Viewing DHCP-Leaned Configuration

Let's see what is automatically configured

```
C:\Users\user>ipconfig /all

[output omitted]

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . : 
    Description . . . . . : Intel(R) 82579LM Gigabit Network Connection
    Physical Address. . . . . : 78-2B-CB-AC-08-67
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 192.168.0.167(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Saturday, January 23, 2021 12:02:04 PM
    Lease Expires . . . . . : Saturday, January 23, 2021 2:02:05 PM
    Default Gateway . . . . . : 192.168.0.1
    DHCP Server . . . . . : 192.168.0.1
    DNS Servers . . . . . : 192.168.0.1
    NetBIOS over Tcpip. . . . . : Enabled

[output omitted]
```

This output shows the configuration learned from DHCP.

- DHCP Enabled: Yes → confirms DHCP usage
- IPv4 address: **192.168.0.167**

Meaning of Preferred:

- The PC previously had this IP address
- When requesting again, it asked for the same IP
- If available, the server assigns it again

Lease Concept (Important):

DHCP uses **leases**, meaning IP addresses are temporary.

1. Lease obtained → when IP was assigned
2. Lease expires → when it must be returned

Key idea:

- IP addresses are **not permanent by default**
- This allows reuse of addresses

Example reasoning: In environments like cafes or airports, many devices connect and disconnect, Lease ensures unused IPs become available again

Basic Function of DHCP

Releasing an IP Address

I want to demonstrate the process that a DHCP client goes through to get an IP address from a DHCP server, but first I need to make my PC released its IP address, so it can ask the server

CLI Output:

```
C:\Users\user>ipconfig /release

Windows IP Configuration

[output omitted]

Ethernet adapter Ethernet0:

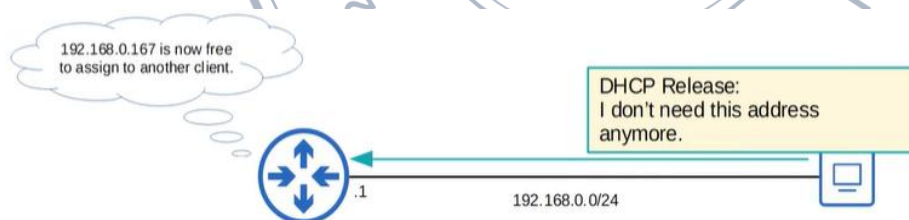
    Connection-specific DNS Suffix  . : 
    Default Gateway . . . . . : 
    . . . . .

[output omitted]
```

Flow Diagram:

PC → DHCP Server "I don't need this IP anymore"

IP 192.168.0.167 → returned to pool



Explanation:

- The PC sends a **DHCP Release message**
- The server marks the IP as available
- It can now be assigned to another client

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Explanation:

1. DHCP uses an **options field** for additional information
2. Option 53 specifies the message type
3. “Magic cookie” is present in DHCP messages

Renewing an IP Address

Now that I will release the IP address of my PC, let's take a look at how my PC can get an IP address from the DHCP servers from the window command prompt I used the command 'ipconfig /renew'

CLI Output:

```
C:\Users\user>ipconfig /renew
C:\Users\user>ipconfig /all

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Description . . . . . : Intel(R) 82579LM Gigabit Network Connection
    Physical Address. . . . . : 78-2B-CB-AC-08-67
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 192.168.0.167(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Saturday, January 23, 2021 3:07:39 PM
    Lease Expires . . . . . : Saturday, January 23, 2021 5:07:38 PM
    Default Gateway . . . . . : 192.168.0.1
    DHCP Server . . . . . : 192.168.0.1
    DNS Servers . . . . . : 192.168.0.1
    NetBIOS over Tcpip. . . . . : Enabled
```

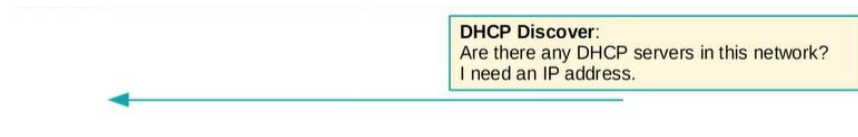
Explanation:

1. The PC contacts the DHCP server again
2. It receives configuration (same or new IP)
3. This process involves four messages, let me introduce them

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DHCP Discover

The first messages are the DHCP Discover message it is a broadcast message from the client



Flow Diagram:

Client → Broadcast

"Are there any DHCP servers?"

"I need an IP address"

Wireshark Packet:

A screenshot of a Wireshark packet capture showing a DHCP Discover message. The packet list shows a frame of 342 bytes on wire (2736 bits) captured on interface \Device\NPF_{9956EC07-3774-4B11-970E}. The packet details pane shows the following information:

No.	Time	Source	Destination	Protocol	Length	Info
261	13:27:34.561617	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xd7alc480

Packet details:

- > Frame 261: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface \Device\NPF_{9956EC07-3774-4B11-970E}
- > Ethernet II, Src: Dell_ac:08:67 (78:2b:cb:ac:08:67), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
- > Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
- > User Datagram Protocol, Src Port: 68, Dst Port: 67
- Dynamic Host Configuration Protocol (Discover)
 - Message type: Boot Request (1)
 - Hardware type: Ethernet (0x01)
 - Hardware address length: 6
 - Hops: 0
 - Transaction ID: 0xd7alc480
 - Seconds elapsed: 0
 - Bootp flags: 0x0000 (Unicast)
 - Client IP address: 0.0.0.0
 - Your (client) IP address: 0.0.0.0
 - Next server IP address: 0.0.0.0
 - Relay agent IP address: 0.0.0.0
 - Client MAC address: Dell_ac:08:67 (78:2b:cb:ac:08:67)
 - Client hardware address padding: 00000000000000000000
 - Server host name not given
 - Boot file name not given
 - Magic cookie: DHCP
 - Option: (53) DHCP Message Type (Discover)
 - Option: (61) Client identifier
 - Option: (50) Requested IP Address (192.168.0.167)
 - Option: (12) Host Name
 - Option: (60) Vendor class identifier
 - Option: (55) Parameter Request List
 - Option: (255) End
 - Padding: 0000000000000000000000000000

Ethernet:

1. Destination MAC: FF:FF:FF:FF:FF:FF

IPv4:

1. Source IP: 0.0.0.0
2. Destination IP: 255.255.255.255

UDP:

1. Source Port: 68
2. Destination Port: 67

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Explanation:

1. Client has no IP address yet → uses source IP 0.0.0.0
2. Destination is broadcast → 255.255.255.255

Reason, Client does not know:

- DHCP server IP
- DHCP server MAC
- Whether a DHCP server even exists

So, it must broadcast

DHCP Message Fields:

1. Client IP Address: 0.0.0.0
2. Client MAC Address: (shown)
3. Bootp Flags: 0000 (unicast)

Important Detail (Bootp Flag):

- Bootp flag = 0000 (unicast)
- Even though message is broadcast:
 - This flag tells server to reply using unicast

DHCP Options:

1. Option 53 → Discover
2. Option 61 → Client Identifier
3. Option 50 → Requested IP Address (192.168.0.167)
 - Client may request previous IP address
 - If available → server may assign same IP
 - If not → different IP is assigned

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DHCP Offer

The Next message is the DHCP offer message, it is sent from the DHCP server to the client, offering an address for the client to use, as well as other information like default gateway, DNS server, etc.



Flow Diagram:

Server → Client

"How about this IP address? + Includes network configuration"

Wireshark Packet:

No.	Time	Source	Destination	Protocol	Length	Info
262	13:27:34.562795	192.168.0.1	192.168.0.167	DHCP	342	DHCP Offer - Transaction ID 0xd7alc480

>	Frame 262: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface \Device\NPF_{9956EC07-3774-4B11-970C}
>	Ethernet II, Src: Tp-LinkT_dd:a8:e4 (98:da:c4:dd:a8:e4), Dst: Dell_ac:08:67 (78:2b:cb:ac:08:67)
>	Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.167
>	User Datagram Protocol, Src Port: 67, Dst Port: 68
>	Dynamic Host Configuration Protocol (Offer)
>	Message type: Boot Reply (2)
>	Hardware type: Ethernet (0x01)
>	Hardware address length: 6
>	Hops: 0
>	Transaction ID: 0xd7alc480
>	Seconds elapsed: 0
>	Bootp flags: 0x0000 (Unicast)
>	Client IP address: 0.0.0.0
>	Your (client) IP address: 192.168.0.167
>	Next server IP address: 192.168.0.1
>	Relay agent IP address: 0.0.0.0
>	Client MAC address: Dell_ac:08:67 (78:2b:cb:ac:08:67)
>	Client hardware address padding: 00000000000000000000
>	Server host name not given
>	Boot file name not given
>	Magic cookie: DHCP
>	Option: (53) DHCP Message Type (Offer)
>	Option: (54) DHCP Server Identifier (192.168.0.1)
>	Option: (51) IP Address Lease Time
>	Option: (58) Renewal Time Value
>	Option: (59) Rebinding Time Value
>	Option: (1) Subnet Mask (255.255.255.0)
>	Option: (28) Broadcast Address (192.168.0.255)
>	Option: (6) Domain Name Server
>	Option: (3) Router
>	Option: (255) End
>	Padding: 0000000000000000

Ethernet:

1. Destination: Client MAC (Unicast)

IPv4:

2. Source: 192.168.0.1
3. Destination: Offered IP (192.168.0.167)

UDP:

1. Source Port: 67
2. Destination Port: 68

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Explanation:

1. Server learned client MAC from Discover:
2. Sends Offer using **unicast**
Offer can be:
 - Unicast
 - OR broadcast
 - Depends on client capability

Bootp Flag Behavior:

1. Client set flag = unicast
→ Server sends unicast But the DHCP offer message can be either broadcast or unicast its depend on the client.
2. In this case my pc told the router to send the offer using unicast, so it did, If my pc told the router to send the offer using broadcast then:
 - The broadcast required:
 - Destination MAC = FF:FF:FF:FF:FF:FF
 - Destination IP = 255.255.255.255

DHCP Message Fields:

Message Type: Offer

Your (Client) IP: 192.168.0.167

DHCP Options

1. Option 53 → Offer
2. Option 54 → DHCP Server Identifier
3. Option 51 → Lease Time
4. Option 58 → Renewal Time
5. Option 59 → Rebinding Time
6. Option 1 → Subnet Mask
7. Option 28 → Broadcast Address
8. Option 6 → DNS Server
9. Option 3 → Default Gateway
10. Option 255 → End

Explanation:

Server provides full configuration:

1. IP address
2. Gateway
3. DNS server
4. Lease duration

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DHCP Request Message

After receiving DHCP Offer messages, the client sends a **DHCP Request** to indicate its selection.



Flow Diagram:

Client → Broadcast

"I want to use the IP address you offered me"

Explanation:

1. Multiple DHCP servers may exist on the same network
2. Each server sends an Offer
3. The client must choose one server

So, the Request message:

- Identifies the selected DHCP server
- Requests to use the offered IP address

Typically: The client accepts the first Offer received

Wireshark Packet:

No.	Time	Source	Destination	Protocol	Length	Info
263	13:27:34.563458	0.0.0.0	255.255.255.255	DHCP	344	DHCP Request - Transaction ID 0xd7alc480

>	Frame 263: 344 bytes on wire (2752 bits), 344 bytes captured (2752 bits) on interface \Device\NPF_{9956EC07-3774-4B11-9700}
>	Ethernet II, Src: Dell_ac:08:67 (78:2b:cb:ac:08:67), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
>	Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
>	User Datagram Protocol, Src Port: 68, Dst Port: 67
>	Dynamic Host Configuration Protocol (Request)
>	Message type: Boot Request (1)
>	Hardware type: Ethernet (0x01)
>	Hardware address length: 6
>	Hops: 0
>	Transaction ID: 0xd7alc480
>	Seconds elapsed: 0
>	Bootp flags: 0x0000 (Unicast)
>	Client IP address: 0.0.0.0
>	Your (client) IP address: 0.0.0.0
>	Next server IP address: 0.0.0.0
>	Relay agent IP address: 0.0.0.0
>	Client MAC address: Dell_ac:08:67 (78:2b:cb:ac:08:67)
>	Client hardware address padding: 00000000000000000000
>	Server host name not given
>	Boot file name not given
>	Magic cookie: DHCP
>	Option: (53) DHCP Message Type (Request)
>	Option: (61) Client identifier
>	Option: (50) Requested IP Address (192.168.0.167)
>	Option: (54) DHCP Server Identifier (192.168.0.1)
>	Option: (12) Host Name
>	Option: (81) Client Fully Qualified Domain Name
>	Option: (60) Vendor class identifier
>	Option: (55) Parameter Request List
>	Option: (255) End

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Ethernet:

1. Destination MAC: FF:FF:FF:FF:FF:FF (Broadcast)

IPv4:

2. Source IP: 0.0.0.0
3. Destination IP: 255.255.255.255

UDP:

1. Source Port: 68
2. Destination Port: 67

Key Details:

1. Sent as broadcast, so all DHCP servers receive it
2. Source IP = 0.0.0.0 (IP not configured yet)
3. Uses UDP:
 - Client → Server = 68 → 67

DHCP Options (Important):

Option 53 → Message Type (Request)

Option 50 → Requested IP Address (192.168.0.167)

Option 54 → DHCP Server Identifier (192.168.0.1)

Option 61 → Client Identifier

Option 255 → End

Understanding the Options:

1. Option 50 → which IP the client wants
2. Option 54 → which server it selected

This is how the client tells:

→ "I am accepting this server's Offer"

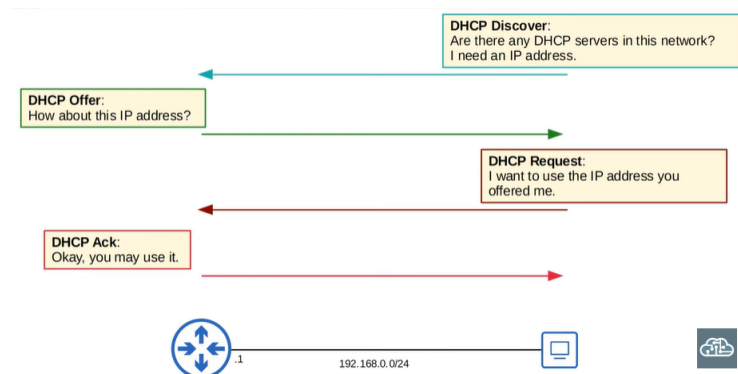
Bootp Flag:

1. Value = 0000 (unicast)
2. Client still sends broadcast
3. But requests server to reply using unicast

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DHCP Ack Message

After receiving the Request, the server sends a **DHCP ACK (Acknowledgement)**.



Flow Diagram:

Server → Client

"Okay, you may use it"

Purpose:

1. Confirms the IP assignment
2. Final step before configuration

After ACK:

→ Client configures the IP address

Wireshark Packet:

No.	Time	Source	Destination	Protocol	Length	Info
264	13:27:34.564862	192.168.0.1	192.168.0.167	DHCP	342	DHCP ACK - Transaction ID 0xd7a1c480

>	Frame 264: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface \Device\NPF_{9956EC07-3774-4B11-97E}
>	Ethernet II, Src: Tp-LinkT_dd:a8:e4 (98:da:c4:dd:a8:e4), Dst: Dell_ac:08:67 (78:2b:cb:ac:08:67)
>	Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.167
>	User Datagram Protocol, Src Port: 67, Dst Port: 68
>	Dynamic Host Configuration Protocol (ACK)
>	Message type: Boot Reply (2)
>	Hardware type: Ethernet (0x01)
>	Hardware address length: 6
>	Hops: 0
>	Transaction ID: 0xd7a1c480
>	Seconds elapsed: 0
>	Bootp flags: 0x0000 (Unicast)
>	Client IP address: 0.0.0.0
>	Your (client) IP address: 192.168.0.167
>	Next server IP address: 192.168.0.1
>	Relay agent IP address: 0.0.0.0
>	Client MAC address: Dell_ac:08:67 (78:2b:cb:ac:08:67)
>	Client hardware address padding: 00000000000000000000
>	Server host name not given
>	Boot file name not given
>	Magic cookie: DHCP
>	Option: (53) DHCP Message Type (ACK)
>	Option: (54) DHCP Server Identifier (192.168.0.1)
>	Option: (51) IP Address Lease Time
>	Option: (58) Renewal Time Value
>	Option: (59) Rebinding Time Value
>	Option: (1) Subnet Mask (255.255.255.0)
>	Option: (28) Broadcast Address (192.168.0.255)
>	Option: (6) Domain Name Server
>	Option: (81) Client Fully Qualified Domain Name
>	Option: (3) Router
>	Option: (255) End
>	Padding: 00

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Ethernet:

1. Destination: Client MAC (Unicast)

IPv4:

2. Source IP: 192.168.0.1
3. Destination IP: 192.168.0.167

UDP:

1. Source Port: 67
2. Destination Port: 68

Key Points:

1. Server → Client communication
2. Uses UDP:
3. Server → Client = 67 → 68
4. Sent as unicast in this case

Important, ACK can be: Unicast OR broadcast (depending on client request)

DHCP Options:

1. Option 53 → ACK
2. Option 54 → DHCP Server Identifier
3. Option 51 → Lease Time [linkedin.com/in/omeshwar23](https://www.linkedin.com/in/omeshwar23)
4. Option 58 → Renewal Time
5. Option 59 → Rebinding Time
6. Option 1 → Subnet Mask
7. Option 28 → Broadcast Address
8. Option 6 → DNS Server
9. Option 3 → Default Gateway
10. Option 255 → End

Explanation:

1. Server provides complete network configuration
2. Includes:
 - IP address
 - Gateway
 - DNS
 - Lease time

Bootp flag again indicates unicast reply

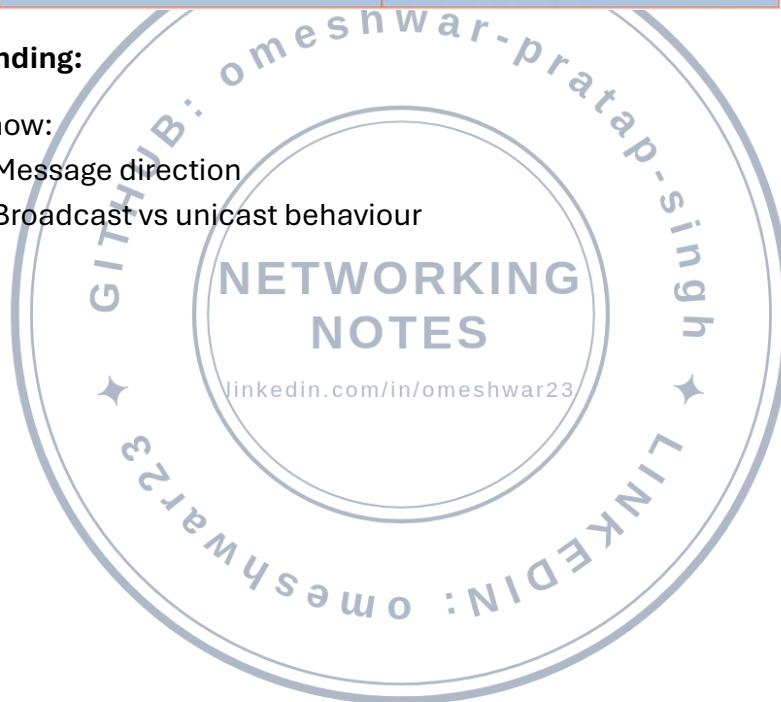
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Full DHCP Process (DORA)

Discover	Client → Server	Broadcast
Offer	Server → Client	Broadcast or Unicast
Request	Client → Server	Broadcast
Ack	Server → Client	Broadcast or Unicast
Release	Client → Server	Unicast

Key Understanding:

1. Must know:
 - Message direction
 - Broadcast vs unicast behaviour



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DHCP Relay

Core Problem:

DHCP Discover is a broadcast message:

Client → 255.255.255.255 (Broadcast)

But:

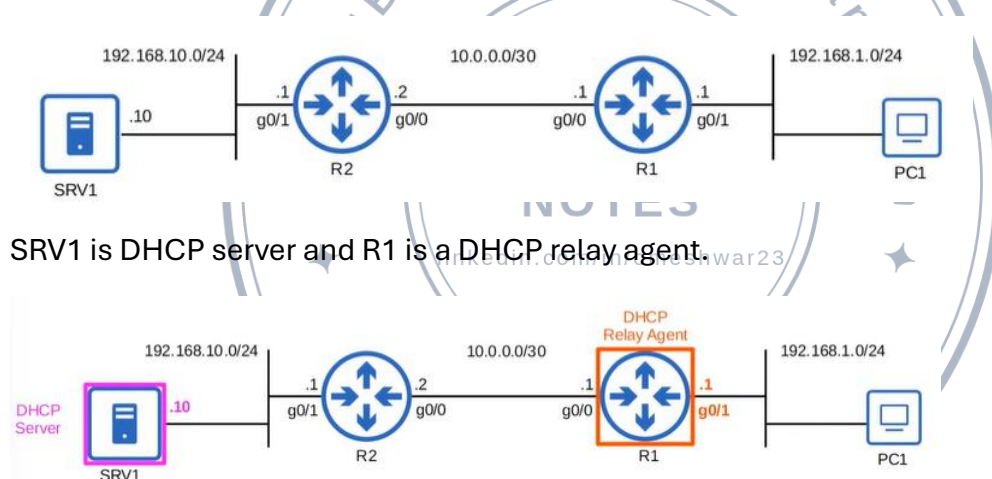
1. Routers do NOT forward broadcast messages
2. So, a DHCP server in another subnet will never receive the request

Solution → DHCP Relay Agent: A router can be configured as a DHCP relay agent.

Its role:

1. Receive client broadcast messages
2. Forward them as unicast messages to the remote DHCP server

Network Topology:



SRV1 is DHCP server and R1 is a DHCP relay agent.

Step-by-Step DHCP Relay Process (Connected Flow):

Step 1: DHCP Discover (Client → Broadcast)

PC1 → Broadcast

Src IP: 0.0.0.0

Dst IP: 255.255.255.255

"Are there any DHCP servers?"

"I need an IP address"

What Happens:

1. PC1 sends broadcast on local network

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2. R1 (relay agent) receives it
3. SRV1 cannot see it directly

Step 2: Relay Agent Forwards Discover (Unicast)

R1 → SRV1 (Unicast)

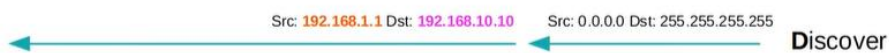
Src IP: 192.168.1.1 (R1 interface)

Dst IP: 192.168.10.10 (DHCP server)

Key Transformation:

1. Broadcast → converted to unicast
2. Source IP changes: → becomes R1's interface IP

Meaning: Server now knows which subnet the request came from



Step 3: DHCP Offer (Server → Relay Agent)

SRV1 → R1

Src IP: 192.168.10.10

Dst IP: 192.168.1.1

What Happens:

1. Server sends Offer to relay agent
2. Includes:
 - IP address (e.g., 192.168.1.100)
 - Subnet mask
 - Gateway
 - DNS

Step 4: Relay Agent Forwards Offer to Client

R1 → PC1

Src IP: 192.168.1.1

Dst: PC1 (Unicast or Broadcast)

Important Detail:

1. R1 forwards the Offer to PC1
2. Can be:
 - Unicast (if possible)
 - OR Broadcast

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Step 5: DHCP Request (Client → Broadcast)

PC1 → Broadcast

"I want to use that IP address"

What Happens:

1. PC1 selects the Offer
2. Broadcasts Request message

Step 6: Relay Agent Forwards Request

R1 → SRV1 (Unicast)

1. Again converts broadcast → unicast
2. Sends to DHCP server



Step 7: DHCP ACK (Server → Relay Agent)

SRV1 → R1

"Okay, you may use it"

Step 8: Relay Agent Forwards ACK to Client

R1 → PC1

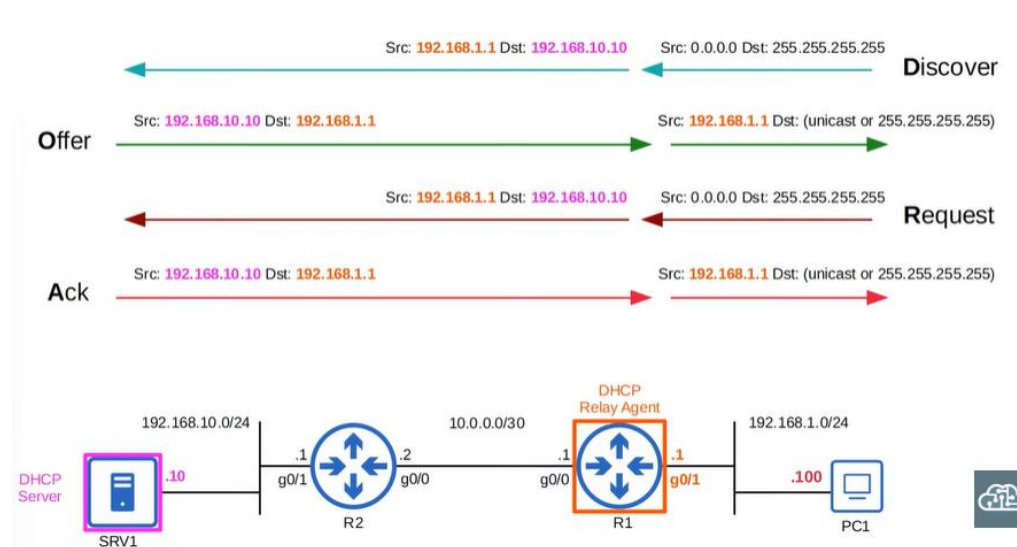
Final Result:

1. PC1 receives ACK
2. Configures IP address
3. Example: IP Address: 192.168.1.100



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Full Diagram:

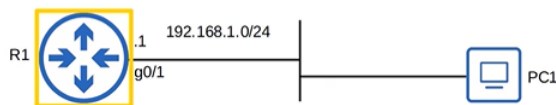


Configuring DHCP in Cisco IOS

Configuring a Cisco Router as a DHCP Server

After understanding DHCP and relay, the next step is configuring a router to act as a DHCP server.

Network Topology:



CLI Configuration

```
R1(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
R1(config)#ip dhcp pool LAB_POOL
R1(dhcp-config)#network 192.168.1.0 ?
/nm or A.B.C.D Network mask or prefix length
<cr>
R1(dhcp-config)#network 192.168.1.0 /24
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#domain-name jeremysitlab.com
R1(dhcp-config)#default-router 192.168.1.1
R1(dhcp-config)#lease 0 5 30
```

Specify a range of addresses that **won't** be given to DHCP clients.

Create a DHCP pool.

Specify the subnet of addresses to be assigned to clients (except the excluded addresses)

Specify the DNS server that DHCP clients should use.

Specify the domain name of the network. (ie. PC1 = pc1.jeremysitlab.com)

Specify the default gateway.

Specify the lease time.
lease *days hours minutes* OR
lease *infinite*

1. Excluding Addresses

```
ip dhcp excluded-address 192.168.1.1 192.168.1.10
```

- Defines a range of IPs NOT given to clients
- Range: 192.168.1.1 → 192.168.1.10

Purpose Reserve IPs for: Routers, Servers, Network devices

Note: This command is optional, but recommended

2. Creating a DHCP Pool

```
ip dhcp pool LAB_POOL
```

- Creates a DHCP pool (logical container)

Definition A DHCP pool =

→ Subnet of assignable IP addresses

→ + configuration (DNS, gateway, etc.)

Important: One pool per network

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3. Defining the Network

network 192.168.1.0 /24

- Specifies subnet for clients

Alternative format: network 192.168.1.0 255.255.255.0

Meaning:

- Router will assign IPs from this subnet
- Except excluded addresses

4. Configuring DNS Server

dns-server 8.8.8.8

- DHCP clients will use this DNS server

Meaning: PC1 will send DNS queries to **8.8.8.8**

5. Configuring Domain Name

domain-name jeremysitlab.com

- Assigns domain to clients

Effect, PC1 becomes part of: pc1.jeremysitlab.com

6. Configuring Default Gateway

default-router 192.168.1.1

- This is the default gateway

Meaning: Clients send traffic outside subnet to R1

7. Configuring Lease Time

lease 0 5 30

Format: lease days hours minutes

Alternative: lease infinite But, Not recommended

Computer Networks - CCNA

DHCP Address Assignment Result (Expected Behaviour)

When PC1 connects:

PC1 → DHCP Discover

R1 → assigns first available IP

Because: .1 → .10 are excluded

So first usable IP = 192.168.1.11

Verifying DHCP on Router

```
R1#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.1.11    0100.0c29.e727.39  Jan 24 2021 10:52 AM    Automatic
```

Explanation

Shows all active DHCP clients

Displays:

- Assigned IP
- Client MAC address
- Lease expiration time
- Binding type

Important: "Automatic" = dynamically assigned

Verifying DHCP on PC

```
C:\Users\user>ipconfig /all

Ethernet adapter Ethernet0:

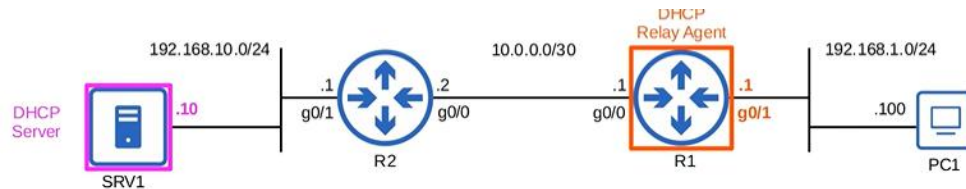
    Connection-specific DNS Suffix  . : jeremysitlab.com
    Description . . . . . : Intel(R) PRO/1000 MT Network Connection #2
    Physical Address. . . . . : 00-0C-29-E7-27-39
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . : Yes
    IPv4 Address. . . . . : 192.168.1.11(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Saturday, January 24, 2021 2:22:35 PM
    Lease Expires . . . . . : Saturday, January 24, 2021 7:52:35 PM
    Default Gateway . . . . . : 192.168.1.1
    DHCP Server . . . . . : 192.168.1.1
    DNS Servers . . . . . : 8.8.8.8
    NetBIOS over Tcpip. . . . . : Enabled
```

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DHCP Relay Agent Configuration

Let's see how to configure DHCP Relay Agent

Network Topology



To configure a router as a DHCP relay agent:

1. Enter interface configuration mode for the interface connected to the client devices.
2. Configure the **IP HELPER-ADDRESS** command with the IP address of the DHCP server.
3. Ensure the router has a route to the DHCP server.
 - If not, configure a static route or use a dynamic routing protocol such as OSPF.

DHCP Relay Agent Configuration on R1

```
R1(config)#interface g0/1
R1(config-if)#ip helper-address 192.168.10.10
R1(config-if)#do show ip interface g0/1
GigabitEthernet0/1 is up, line protocol is up
Internet address is 192.168.1.1/24
Broadcast address is 255.255.255.255
Address determined by non-volatile memory
MTU is 1500 bytes
Helper address is 192.168.10.10
[output omitted]
```

Configure the interface connected to the subnet of the client devices.

Configure the IP address of the DHCP server as the 'helper' address.

Explanation:

Configure the interface connected to the client subnet (**G0/1**).

Use: `ip helper-address 192.168.10.10`

The configured helper address is the IP address of **SRV1**, the DHCP server.

Verification with: `show ip interface g0/1`

shows: Helper address is 192.168.10.10

That is all that is required to configure a router as a DHCP relay agent:

`ip helper-address <DHCP-server-IP>`

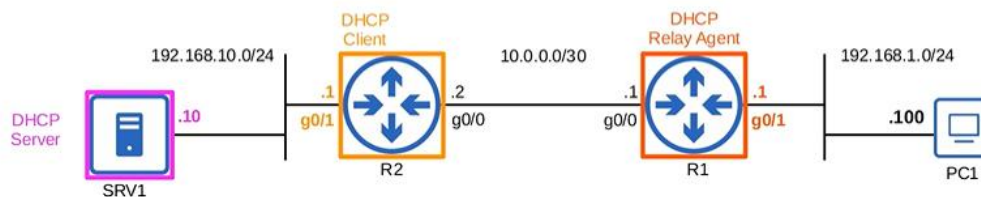
Computer Networks - CCNA

Configuring a Cisco Router as a DHCP Client

A Cisco router can act as a **DHCP client** and obtain an IP address for one of its interfaces using DHCP.

Although network devices are usually configured with fixed IP addresses, this feature can be configured when needed.

Network Topology



In this configuration, **R2's G0/1 interface** will operate as a DHCP client.

DHCP Client Configuration on R2

```
R2(config)#interface g0/1
R2(config-if)#ip address dhcp
R2(config-if)#do sh ip interface g0/1
GigabitEthernet0/1 is up, line protocol is up
Internet address is 192.168.10.1/24
Broadcast address is 255.255.255.255
Address determined by DHCP
[output omitted]
```

Use the **ip address dhcp** mode to tell the router to use DHCP to learn its IP address.

Explanation

From interface configuration mode on **G0/1**, configure: **ip address dhcp**

This causes R2 to:

1. Broadcast a DHCP Discover message.
2. Obtain an IP address from **SRV1**.

Verification with: **show ip interface g0/1**

Displays: Address determined by DHCP confirming that the interface received its address through DHCP.

Note: Only **G0/1** is operating as a DHCP client. **G0/0** remains manually configured.

Computer Networks - CCNA

Commands Covered

The lesson covered:

- Windows command prompt DHCP commands.
- Cisco IOS commands to configure:
 - A DHCP server.
 - A DHCP relay agent.
 - A DHCP client.

```
C:\Users\user> ipconfig /release
```

```
C:\Users\user> ipconfig /renew
```

```
R1(config)# ip dhcp excluded-address low-address high-address
```

```
R1(config)# ip dhcp pool pool-name
```

```
R1(dhcp-config)# network ip-address {/prefix-length | subnet-mask}
```

```
R1(dhcp-config)# dns-server ip-address
```

```
R1(dhcp-config)# domain-name domain-name
```

```
R1(dhcp-config)# default-router ip-address
```

```
R1(dhcp-config)# lease {days hours minutes | infinite}
```

```
R1# show ip dhcp binding
```

```
R1(config-if)# ip helper-address ip-address
```

DHCP relay agent

```
R1(config-if)# ip address dhcp
```

DHCP client

DHCP server

